Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: 3A Teixeira

Physical address of dairy:

Number and Street	City	County	Zip Code
12926 N Vincent RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X041-X130-X049-XXXX

B. OPERATORS

Teixeira, Antonio

Operator name: Teixeira, Antonio

Telephone no.: (209) 968-0610 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5200 Karyn Dawn DR	Denair	CA	95316

C. OWNERS

Fiorini, Victor D

Legal owner name: Fiorini, Victor D

Telephone no.: (209) 632-6935 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12926 N Vincent RD	Turlock	CA	95380

This owner is responsible for paying permit fees.

Pontes, Anthony J

Legal owner name: Pontes, Anthony J

Telephone no.: (209) 632-5715 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
13094 N Vincent RD	Turlock	CA	95380

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	190	235	110	0
Number under roof	826	140	0	0	0	0
Maximum number	826	140	190	235	110	0
Average number	826	140	190	235	110	0
Avg live weight (lbs)	1,300	1,350	900	600		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 27,630.38 *tons per reporting period*

Total nitrogen from manure: 348,527.46 *lbs per reporting period*

After ammonia losses (30% loss applied): 243,969.22 *lbs per reporting period*

Total phosphorus from manure: 57,494.08 *lbs per reporting period*

Total potassium from manure: 158,186.62 *lbs per reporting period*

Total salt from manure: 421,115.10 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 19,528,375 *gallons*

Total nitrogen generated: 107,585.08 *lbs*

Total phosphorus generated: 7,939.51 *lbs*

Total potassium generated: 116,686.18 *lbs*

Total salt generated: 983,405.78 *lbs*

	19,528,375	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	19,528,375	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Dairy Deep Well	Ground water
Ranch Deep Well	Ground water
TID Highline Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
04/24/2023	Solid commercial fertilizer *UN32*	6.00 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
06/15/2023	Solid commercial fertilizer *4-10-10*	20.70 *ton*	Dry-weight	0.1	4.000000	4.400000	8.300000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	5,490.50	1,819.78	3,432.76	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	5,490.50	1,819.78	3,432.76	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
field 10	11	11	2	process wastewater	X041-X130-X049-XXXX
field 11-12	16	16	2	manure	X041-X130-X048-XXXX
field 13	37	37	2	process wastewater	X041-X130-X003-XXXX
field 14	6	6	1	process wastewater	X041-X130-X049-XXXX
field 2	55	55	2	process wastewater	X024-X009-X025-XXXX
field 3	40	40	1	manure	X024-X009-X025-XXXX X041-X080-X003-XXXX
field 4	30	30	1	process wastewater	X024-X009-X025-XXXX X041-X080-X003-XXXX
field 5	17	17	1	process wastewater	X041-X080-X003-XXXX
field 6-7	37	37	2	process wastewater	X041-X080-X008-XXXX
field 8	36	36	2	process wastewater	X041-X080-X003-XXXX
field 9	15	15	2	process wastewater	X041-X130-X049-XXXX
Walnuts	36	36	1	process wastewater	X024-X009-X018-XXXX X041-X140-X009-XXXX X041-X140-X010-XXXX
Totals for areas that were used for application	336	336	19		
Totals for areas that were not used for application					
Land application area totals	336	336	19		

B. CROPS AND HARVESTS

field 10

Field name: field 10

field 10

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	142.00 *ton*	As-is		59.4	7,230.00	1,710.00	21,100.00		9.12

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.91	186.67	44.15	544.76	955.97

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	224.00 *ton*	As-is		71.8	3,950.00	910.00	7,380.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	20.36	160.87	37.06	300.57	735.05

field 11-12

Field name: field 11-12

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	298.00 *ton*	As-is		68.4	3,860.00	1,110.00	10,100.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.63	143.79	41.35	376.23	906.37

field 11-12

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	337.00 *ton*	As-is		65.0	3,890.00	1,020.00	3,840.00		63.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	21.06	163.87	42.97	161.76	9,332.79

field 13

Field name: field 13

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	491.00 *ton*	As-is		68.6	7,440.00	1,600.00	17,300.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	13.27	197.46	42.46	459.15	908.38

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	608.00 *ton*	As-is		71.0	5,190.00	950.00	7,540.00		8.03

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	16.43	170.57	31.22	247.80	765.32

field 14

Field name: field 14

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 6 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	69.00 *ton*	As-is		67.4	8,020.00	1,500.00	17,300.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	11.50	184.46	34.50	397.90	899.76

field 2

Field name: field 2

11/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	1,042.00 *ton*	As-is		68.8	3,380.00	820.00	5,100.00		9.52

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	18.95	128.07	31.07	193.24	1,125.45

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	963.00 *ton*	As-is		64.2	4,830.00	1,020.00	5,530.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	17.51	169.14	35.72	193.65	739.65

field 3

Field name: field 3

12/18/2017: Almond, in shell

Crop: Almond, in shell Acres planted: 40 Plant date: 12/18/2017

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	163.00 *ton*	As-is		7.8	24,200.00	1,700.00	15,900.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.08	197.23	13.86	129.59	879.17

field 4

Field name: field 4

11/04/2021: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 30 Plant date: 11/04/2021

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	212.00 *ton*	As-is		9.5	26,200.00	1,650.00	16,500.00		15.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	7.07	370.29	23.32	233.20	1,982.55

field 5

Field name: field 5

field 5

10/22/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 17 Plant date: 10/22/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	107.00 *ton*	As-is		10.2	28,100.00	1,810.00	17,500.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	17.00	1,020.00	91.80	714.00	0.00
Total actual harvest content	6.29	353.73	22.78	220.29	1,322.60

field 6-7

Field name: field 6-7

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	621.00 *ton*	As-is		67.6	4,570.00	1,040.00	12,000.00		9.24

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	16.78	153.40	34.91	402.81	1,004.93

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/11/2023	676.00 *ton*	As-is		67.0	4,690.00	1,170.00	6,240.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	18.27	171.38	42.75	228.01	711.44

field 8

Field name: field 8

11/07/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/07/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/19/2023	703.00 *ton*	As-is		65.4	3,860.00	790.00	5,660.00		8.07

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	19.53	150.75	30.85	221.05	1,090.52

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	703.00 *ton*	As-is		73.2	4,150.00	740.00	5,810.00		8.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	19.53	162.08	28.90	226.91	921.09

field 9

Field name: field 9

11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/06/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	194.00 *ton*	As-is		68.4	7,300.00	1,490.00	17,400.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.93	188.83	38.54	450.08	989.04

field 9

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	255.00 *ton*	As-is		61.8	4,800.00	1,000.00	6,100.00		2.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	25.00	200.00	37.50	165.00	0.00
Total actual harvest content	17.00	163.20	34.00	207.40	298.72

Walnuts

Field name: Walnuts

01/01/2008: Walnuts

Crop: Walnuts Acres planted: 36 Plant date: 01/01/2008

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	71.00 *ton*	As-is		7.0	38,600.00	4,420.00	5,780.00		1.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	2.50	100.00	7.50	72.00	0.00
Total actual harvest content	1.97	152.26	17.43	22.80	65.30

NUTRIENT BUDGET

A. LAND APPLICATIONS

field 10 - 11/06/2022: Oats, silage-soft dough

Field name: field 10

Crop: Oats, silage-soft dough — Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.66	8.38	96.57	515.63	192,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	1,620,000.00 *gal*
Application event totals		78.66	8.38	96.57	515.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	180.25	12.89	66.42	408.57	144,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	30.42	1,215,000.00 *gal*
Application event totals		180.25	12.89	66.42	438.99	

field 10 - 06/15/2023: Corn, silage

Field name: field 10

Crop: Corn, silage — Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.06	4.79	141.65	1,620.45	240,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	43.94	1,755,000.00 *gal*
Application event totals		111.06	4.79	141.65	1,664.38	

field 10 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
Application event totals		8.00	8.80	16.60	0.00	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	35.15	1,404,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	49.30	5.39	90.13	606.30	180,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	32.95	1,316,250.00 *gal*
Application event totals		49.30	5.39	90.13	639.25	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	35.15	1,404,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.15	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.01	5.03	84.12	565.88	168,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	30.76	1,228,500.00 *gal*
Application event totals		46.01	5.03	84.12	596.64	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	30.76	1,228,500.00 *gal*
Application event totals		0.00	0.00	0.00	30.76	

field 10 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	30.76	1,228,500.00 *gal*
Application event totals		0.00	0.00	0.00	30.76	

field 11-12 - 11/06/2022: Oats, silage-soft dough

Field name: field 11-12

Crop: Oats, silage-soft dough Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	195.08	29.37	70.21	1,264.81	350.00 *ton*
Application event totals		195.08	29.37	70.21	1,264.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	47.40	2,754,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.40	

field 11-12 - 06/14/2023: Corn, silage

Field name: field 11-12

Crop: Corn, silage Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

field 11-12 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	206.24	92.91	385.19	5,651.49	120.00 *ton*
Application event totals		206.24	92.91	385.19	5,651.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	2,592,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
Application event totals		8.00	8.80	16.60	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	41.82	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	40.43	2,349,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	41.82	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.82	

field 11-12 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	39.04	2,268,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	36.25	2,106,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	36.25	2,106,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.25	

field 13 - 11/07/2022: Oats, silage-soft dough

Field name: field 13

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.46	6.22	71.78	383.24	480,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	4,320,000.00 *gal*
Application event totals		58.46	6.22	71.78	383.24	

field 13 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	178.63	12.77	65.82	404.89	480,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	32.15	4,320,000.00 *gal*
Application event totals		178.63	12.77	65.82	437.04	

field 13 - 06/14/2023: Corn, silage

Field name: field 13

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
WWS	Process wastewater	115.57	4.98	147.40	1,686.14	840,000.00 *gal*
Application event totals		123.57	13.78	164.00	1,686.14	

06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	45.01	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.01	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	41.80	5,616,000.00 *gal*
Application event totals		0.00	0.00	0.00	41.80	

field 13 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.74	5.88	98.25	660.92	660,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	35.37	4,752,000.00 *gal*
Application event totals		53.74	5.88	98.25	696.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	38.58	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.74	5.88	98.25	660.92	660,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	35.37	4,752,000.00 *gal*
Application event totals		53.74	5.88	98.25	696.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	35.37	4,752,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	32.15	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	32.15	

field 14 - 11/07/2022: Oats, silage-soft dough

Field name: field 14

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

field 14 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/22/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.59	7.20	82.99	443.12	90,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	810,000.00 *gal*
Application event totals		67.59	7.20	82.99	443.12	

03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	172.12	12.31	63.42	390.13	75,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	30.98	675,000.00 *gal*
Application event totals		172.12	12.31	63.42	421.11	

field 2 - 11/08/2022: Oats, silage-soft dough

Field name: field 2

Crop: Oats, silage-soft dough

Plant date: 11/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	171.44	14.96	74.32	220.00	653,125.00 *gal*
Application event totals		171.44	14.96	74.32	220.00	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	50.70	10,125,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.70	

field 2 - 06/15/2023: Corn, silage

field 2 - 06/15/2023: Corn, silage

Field name: field 2

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.62	4.60	135.99	1,555.63	1,152,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	48.67	9,720,000.00 *gal*
Application event totals		106.62	4.60	135.99	1,604.30	

06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
Application event totals		8.00	8.80	16.60	0.00	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.61	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.21	6.04	100.94	679.06	1,008,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	42.58	8,505,000.00 *gal*
Application event totals		55.21	6.04	100.94	721.64	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	8,910,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.61	

field 2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.58	5.75	96.13	646.72	960,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	40.56	8,100,000.00 *gal*
Application event totals		52.58	5.75	96.13	687.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	38.53	7,695,000.00 *gal*
Application event totals		0.00	0.00	0.00	38.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	36.50	7,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	36.50	

field 3 - 12/18/2017: Almond, in shell

Field name: field 3

Crop: Almond, in shell

Plant date: 12/18/2017

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	3,859,269.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/12/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separated Solids	Separator solids	46.82	7.05	16.85	303.56	210.00 *ton*
Application event totals		46.82	7.05	16.85	303.56	

field 3 - 12/18/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	7,718,538.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	96.00	0.00	0.00	0.00	
Application event totals		96.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	7,718,538.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	7,718,538.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	7,718,538.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	7,718,538.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

field 3 - 12/18/2017: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	3,859,269.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

field 4 - 11/04/2021: Alfalfa, hay

Field name: field 4

Crop: Alfalfa, hay

Plant date: 11/04/2021

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	376.69	16.24	480.44	5,496.02	2,220,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	45.85	4,995,000.00 *gal*
Application event totals		376.69	16.24	480.44	5,541.87	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.61	

06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	43.37	4,725,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.37	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	42.13	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.13	

field 4 - 11/04/2021: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	40.89	4,455,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	39.66	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	4,860,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.61	

field 5 - 10/22/2018: Alfalfa, hay

Field name: field 5

Crop: Alfalfa, hay

Plant date: 10/22/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	395.26	17.04	504.12	5,766.89	1,320,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	48.11	2,970,000.00 *gal*
Application event totals		395.26	17.04	504.12	5,815.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	50.30	3,105,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.30	

field 5 - 10/22/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	48.11	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	48.11	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	48.11	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	48.11	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	45.92	2,835,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.92	

field 6-7 - 11/06/2022: Oats, silage-soft dough

Field name: field 6-7

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

field 6-7 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	205.73	17.96	89.19	263.99	527,250.00 *gal*
Application event totals		205.73	17.96	89.19	263.99	

04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	45.22	6,075,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.22	

field 6-7 - 06/15/2023: Corn, silage

Field name: field 6-7

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.57	4.98	147.40	1,686.14	840,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	47.83	6,426,000.00 *gal*
Application event totals		115.57	4.98	147.40	1,733.97	

06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
Application event totals		8.00	8.80	16.60	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.41	5,967,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.41	

field 6-7 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.62	6.41	107.18	721.01	720,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	41.00	5,508,000.00 *gal*
Application event totals		58.62	6.41	107.18	762.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.41	5,967,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.18	6.15	102.71	690.97	690,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	39.29	5,278,500.00 *gal*
Application event totals		56.18	6.15	102.71	730.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	37.58	5,049,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	37.58	5,049,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.58	

field 8 - 11/07/2022: Oats, silage-soft dough

Field name: field 8

Crop: Oats, silage-soft dough

Plant date: 11/07/2022

field 8 - 11/07/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/20/2022	Towed tank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater sludge	205.73	17.96	89.19	263.99	513,000.00 *gal*
Application event totals		205.73	17.96	89.19	263.99	

04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	47.09	6,156,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.09	

field 8 - 06/14/2023: Corn, silage

Field name: field 8

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	101.81	4.39	129.85	1,485.41	720,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	47.09	6,156,000.00 *gal*
Application event totals		101.81	4.39	129.85	1,532.50	

06/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
Application event totals		8.00	8.80	16.60	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	5,832,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.61	

field 8 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.25	6.59	110.15	741.04	720,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	43.37	5,670,000.00 *gal*
Application event totals		60.25	6.59	110.15	784.41	

08/12/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	43.37	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.37	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.20	5.27	88.12	592.83	576,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	42.75	5,589,000.00 *gal*
Application event totals		48.20	5.27	88.12	635.58	

09/07/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	42.13	5,508,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.13	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	40.28	5,265,000.00 *gal*
Application event totals		0.00	0.00	0.00	40.28	

field 9 - 11/06/2022: Oats, silage-soft dough

Field name: field 9

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

field 9 - 11/06/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	72.10	7.68	88.52	472.66	240,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	2,160,000.00 *gal*
Application event totals		72.10	7.68	88.52	472.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	165.23	11.82	60.89	374.52	180,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	29.74	1,620,000.00 *gal*
Application event totals		165.23	11.82	60.89	404.27	

field 9 - 06/29/2023: Corn, silage

Field name: field 9

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	111.99	4.83	142.83	1,633.95	330,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	54.53	2,970,000.00 *gal*
Application event totals		111.99	4.83	142.83	1,688.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-10-10	Solid commercial fertilizer	8.00	8.80	16.60	0.00	
Application event totals		8.00	8.80	16.60	0.00	

field 9 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	47.09	2,565,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.23	5.93	99.14	666.93	270,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	44.61	2,430,000.00 *gal*
Application event totals		54.23	5.93	99.14	711.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	39.66	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.20	5.27	88.12	592.83	240,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	39.66	2,160,000.00 *gal*
Application event totals		48.20	5.27	88.12	632.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	42.13	2,295,000.00 *gal*
Application event totals		0.00	0.00	0.00	42.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	39.66	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	39.66	

Walnuts - 01/01/2008: Walnuts

Field name: Walnuts

Crop: Walnuts

Plant date: 01/01/2008

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/31/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.64	10.40	119.88	640.06	780,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	0.00	5,967,000.00 *gal*
Application event totals		97.64	10.40	119.88	640.06	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.32	4.02	119.03	1,361.63	660,000.00 *gal*
TID Highline Canal	Surface water	0.00	0.00	0.00	38.62	5,049,000.00 *gal*
Application event totals		93.32	4.02	119.03	1,400.25	

05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	35.11	4,590,000.00 *gal*
Application event totals		0.00	0.00	0.00	35.11	

05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	31.60	4,131,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.60	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	31.60	4,131,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.60	

Walnuts - 01/01/2008: Walnuts

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	31.60	4,131,000.00 *gal*
Application event totals		0.00	0.00	0.00	31.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	28.09	3,672,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	28.09	3,672,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	28.09	3,672,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	28.09	3,672,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.09	

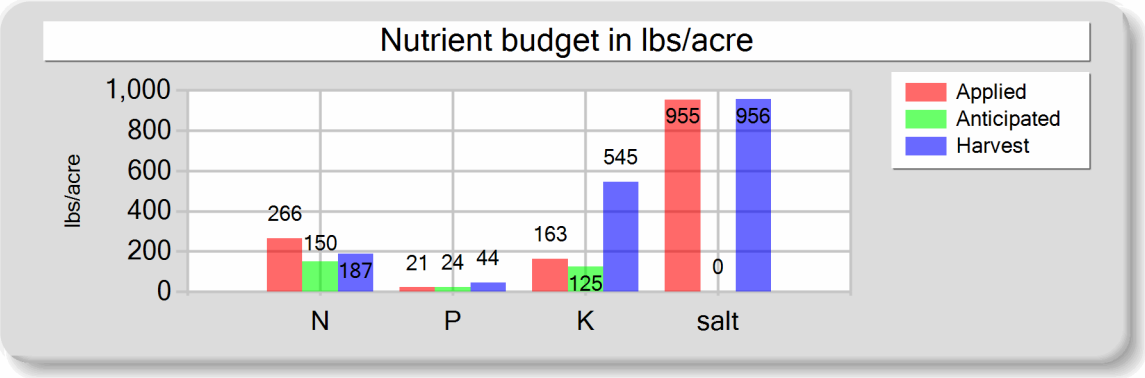
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Highline Canal	Surface water	0.00	0.00	0.00	28.09	3,672,000.00 *gal*
Application event totals		0.00	0.00	0.00	28.09	

B. NUTRIENT BUDGET

field 10 - 11/06/2022: Oats, silage-soft dough

Field name: field 10 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	258.91	21.27	162.99	924.20
Fresh water	0.00	0.00	0.00	30.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.91	21.27	162.99	954.62
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	186.67	44.15	544.76	955.97
Nutrient balance	79.24	-22.88	-381.77	-1.36
Applied to removed ratio	1.42	0.48	0.30	1.00

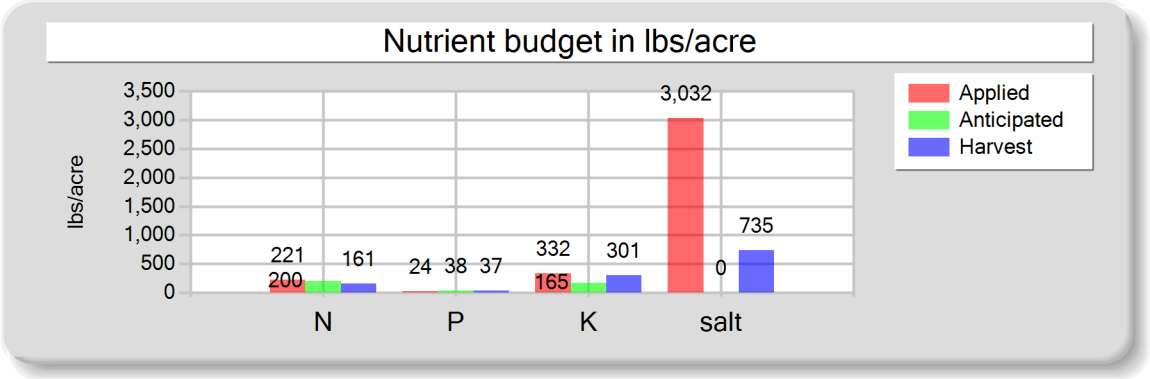
Fresh water applied	
2,835,000.00	*gallons*
104.40	*acre-inches*
9.49	*inches/acre*

Process wastewater applied	
336,000.00	*gallons*
12.37	*acre-inches*
1.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 10 - 06/15/2023: Corn, silage

Field name: field 10 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.37	15.22	315.90	2,792.63
Fresh water	0.00	0.00	0.00	239.45
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.37	24.02	332.50	3,032.09
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	160.87	37.06	300.57	735.05
Nutrient balance	60.50	-13.05	31.93	2,297.04
Applied to removed ratio	1.38	0.65	1.11	4.13

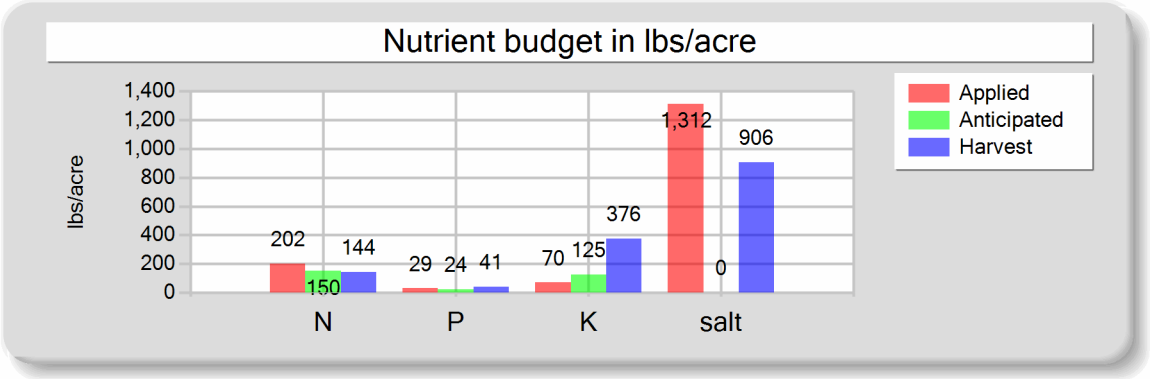
Fresh water applied	
9,564,750.00	*gallons*
352.24	*acre-inches*
32.02	*inches/acre*

Process wastewater applied	
588,000.00	*gallons*
21.65	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 11-12 - 11/06/2022: Oats, silage-soft dough

Field name: field 11-12 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	195.08	29.37	70.21	1,264.81
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	47.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.08	29.37	70.21	1,312.21
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	143.79	41.35	376.23	906.37
Nutrient balance	58.30	-11.98	-306.02	405.85
Applied to removed ratio	1.41	0.71	0.19	1.45

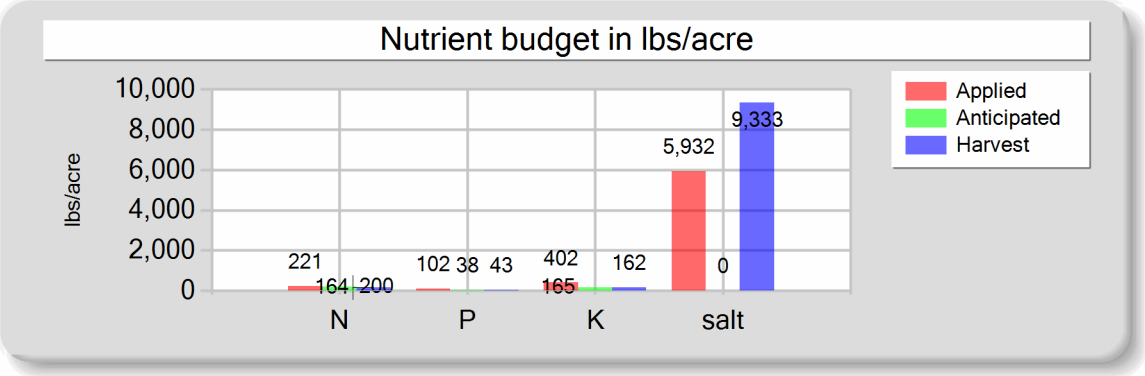
Fresh water applied	
2,754,000.00	*gallons*
101.42	*acre-inches*
6.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 11-12 - 06/14/2023: Corn, silage

Field name: field 11-12 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	206.24	92.91	385.19	5,651.49
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	280.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.24	101.71	401.79	5,931.71
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	163.87	42.97	161.76	9,332.79
Nutrient balance	57.38	58.74	240.03	-3,401.08
Applied to removed ratio	1.35	2.37	2.48	0.64

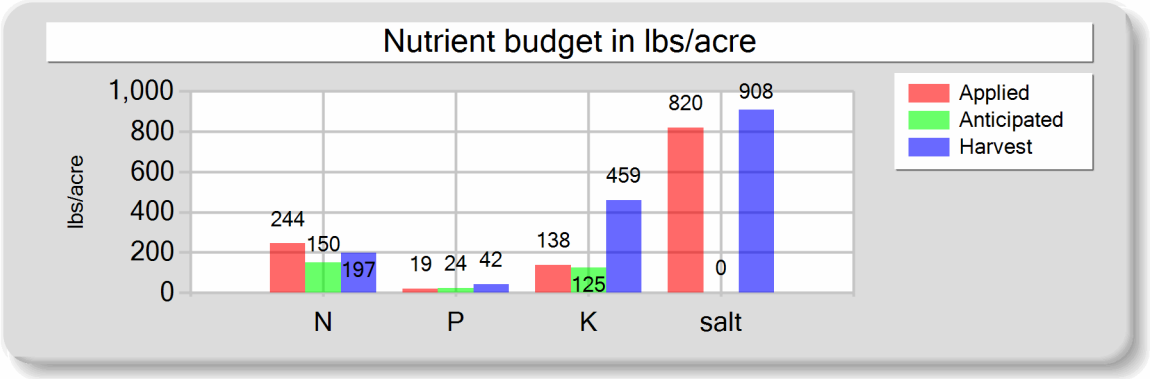
Fresh water applied	
16,281,000.00	*gallons*
599.57	*acre-inches*
37.47	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 13 - 11/07/2022: Oats, silage-soft dough

Field name: field 13 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	237.09	19.00	137.60	788.13
Fresh water	0.00	0.00	0.00	32.15
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.09	19.00	137.60	820.28
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	197.46	42.46	459.15	908.38
Nutrient balance	46.63	-23.47	-321.55	-88.09
Applied to removed ratio	1.24	0.45	0.30	0.90

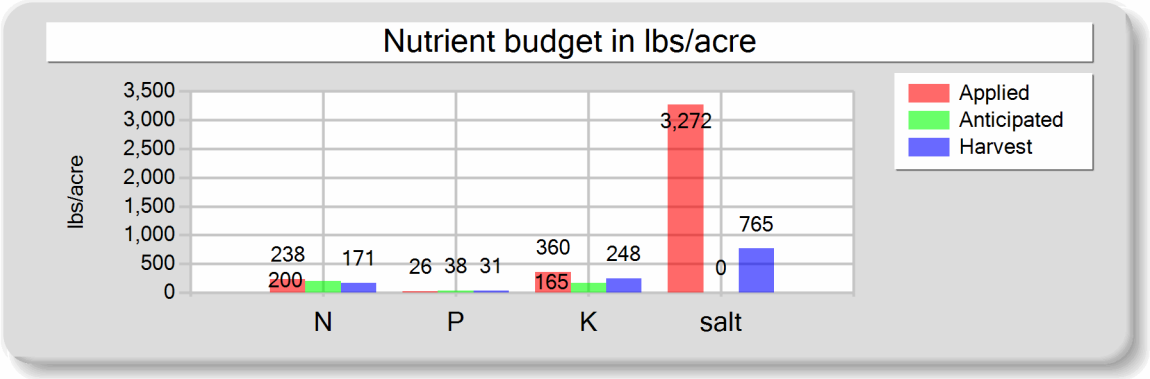
Fresh water applied	
8,640,000.00	*gallons*
318.18	*acre-inches*
8.60	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 13 - 06/14/2023: Corn, silage

Field name: field 13 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	223.04	16.74	343.89	3,007.99
Fresh water	0.00	0.00	0.00	263.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.04	25.54	360.49	3,271.64
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	170.57	31.22	247.80	765.32
Nutrient balance	67.47	-5.68	112.69	2,506.32
Applied to removed ratio	1.40	0.82	1.45	4.27

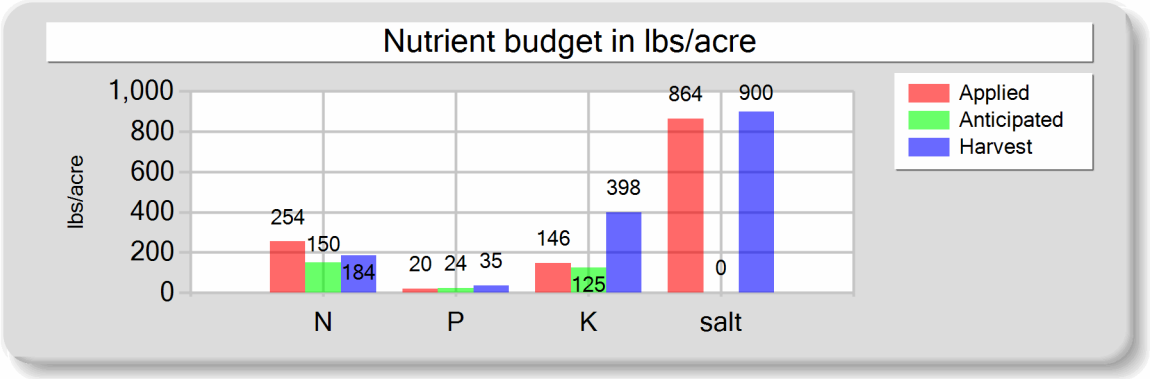
Fresh water applied	
35,424,000.00	*gallons*
1,304.55	*acre-inches*
35.26	*inches/acre*

Process wastewater applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
2.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 14 - 11/07/2022: Oats, silage-soft dough

Field name: field 14 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.71	19.51	146.41	833.25
Fresh water	0.00	0.00	0.00	30.98
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	253.71	19.51	146.41	864.23
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	184.46	34.50	397.90	899.76
Nutrient balance	69.25	-14.99	-251.49	-35.53
Applied to removed ratio	1.38	0.57	0.37	0.96

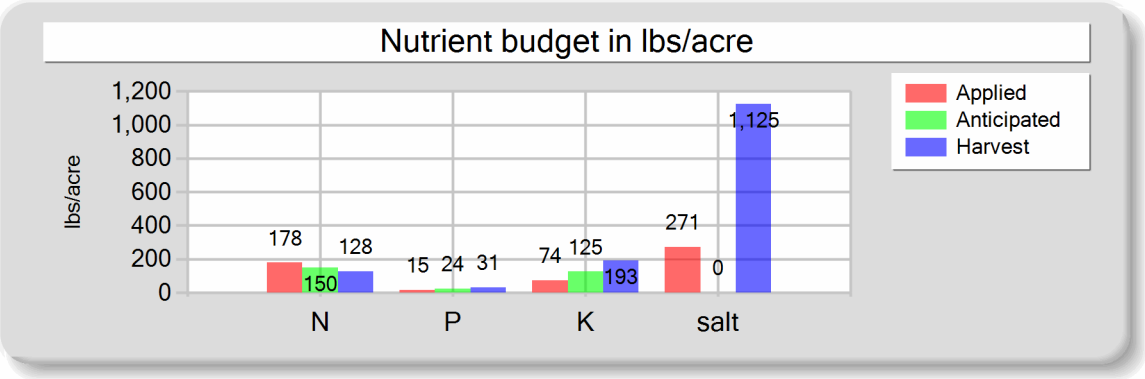
Fresh water applied	
1,485,000.00	*gallons*
54.69	*acre-inches*
9.11	*inches/acre*

Process wastewater applied	
165,000.00	*gallons*
6.08	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 2 - 11/08/2022: Oats, silage-soft dough

Field name: field 2 Crop: Oats, silage-soft dough Plant date: 11/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.44	14.96	74.32	220.00
Fresh water	0.00	0.00	0.00	50.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	178.44	14.96	74.32	270.69
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	128.07	31.07	193.24	1,125.45
Nutrient balance	50.37	-16.11	-118.92	-854.76
Applied to removed ratio	1.39	0.48	0.38	0.24

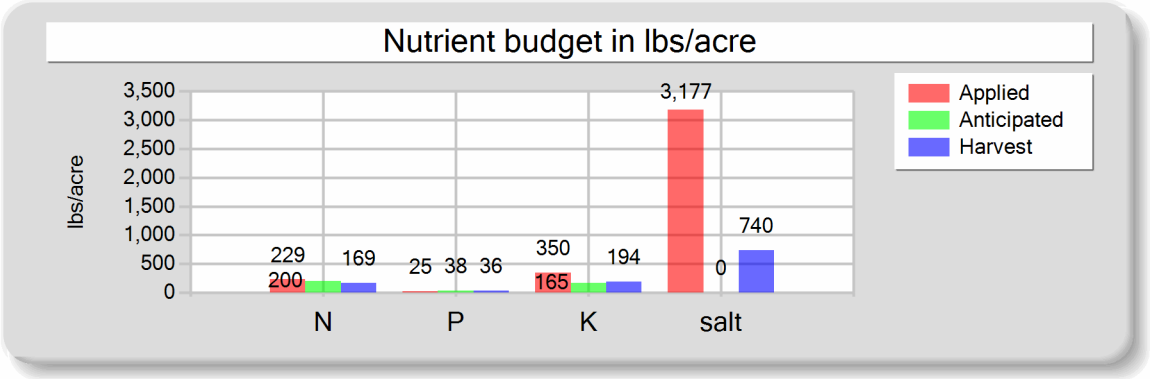
Fresh water applied	
10,125,000.00	*gallons*
372.87	*acre-inches*
6.78	*inches/acre*

Process wastewater applied	
653,125.00	*gallons*
24.05	*acre-inches*
0.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 2 - 06/15/2023: Corn, silage

Field name: field 2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.42	16.39	333.06	2,881.41
Fresh water	0.00	0.00	0.00	296.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.42	25.19	349.66	3,177.47
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	169.14	35.72	193.65	739.65
Nutrient balance	60.28	-10.53	156.01	2,437.82
Applied to removed ratio	1.36	0.71	1.81	4.30

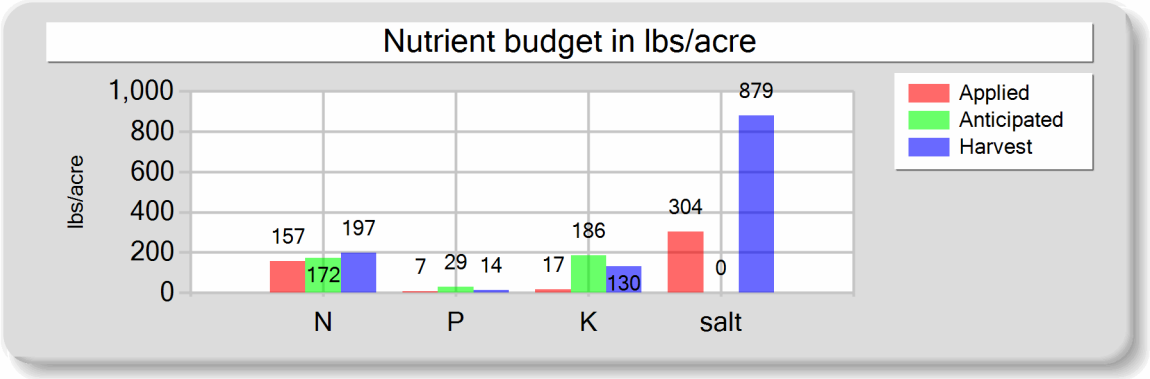
Fresh water applied	
59,130,000.00	*gallons*
2,177.56	*acre-inches*
39.59	*inches/acre*

Process wastewater applied	
3,120,000.00	*gallons*
114.90	*acre-inches*
2.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 3 - 12/18/2017: Almond, in shell

Field name: field 3 Crop: Almond, in shell Plant date: 12/18/2017



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	96.00	0.00	0.00	0.00
Dry manure	46.82	7.05	16.85	303.56
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	156.82	7.05	16.85	303.56
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	197.23	13.86	129.59	879.17
Nutrient balance	-40.41	-6.81	-112.74	-575.62
Applied to removed ratio	0.80	0.51	0.13	0.35

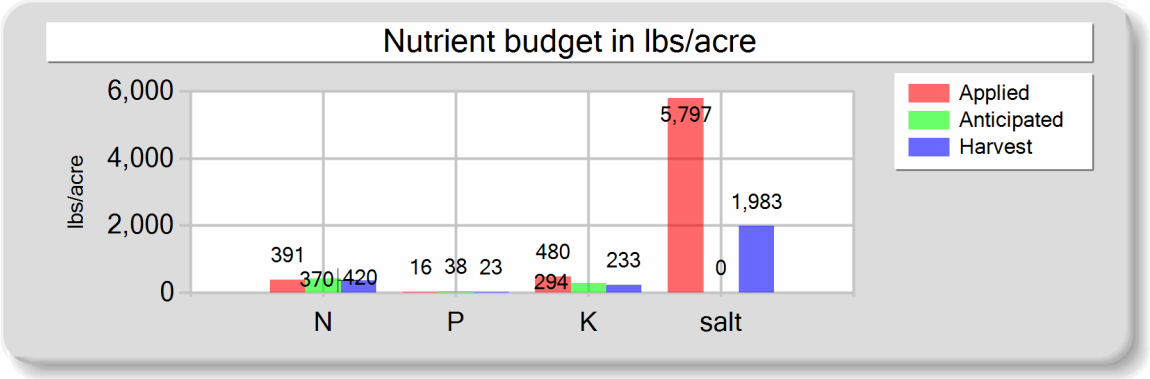
Fresh water applied	
46,311,228.00	*gallons*
1,705.49	*acre-inches*
42.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 4 - 11/04/2021: Alfalfa, hay

Field name: field 4 Crop: Alfalfa, hay Plant date: 11/04/2021



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	376.69	16.24	480.44	5,496.02
Fresh water	0.00	0.00	0.00	301.13
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	390.69	16.24	480.44	5,797.15
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	370.29	23.32	233.20	1,982.55
Nutrient balance	20.40	-7.08	247.24	3,814.60
Applied to removed ratio	1.06	0.70	2.06	2.92

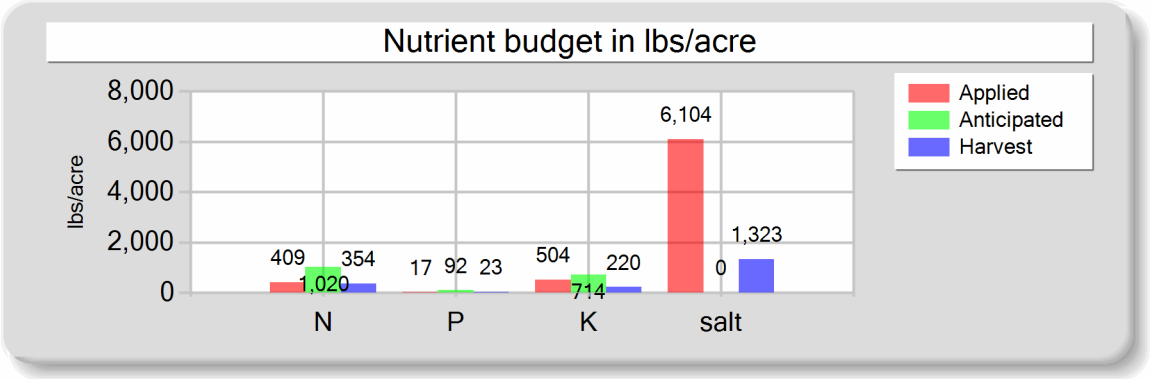
Fresh water applied	
32,805,000.00	*gallons*
1,208.10	*acre-inches*
40.27	*inches/acre*

Process wastewater applied	
2,220,000.00	*gallons*
81.76	*acre-inches*
2.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 5 - 10/22/2018: Alfalfa, hay

Field name: field 5 Crop: Alfalfa, hay Plant date: 10/22/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	395.26	17.04	504.12	5,766.89
Fresh water	0.00	0.00	0.00	336.78
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	409.26	17.04	504.12	6,103.67
Anticipated crop nutrient removal	1,020.00	91.80	714.00	0.00
Actual crop nutrient removal	353.73	22.78	220.29	1,322.60
Nutrient balance	55.53	-5.74	283.82	4,781.07
Applied to removed ratio	1.16	0.75	2.29	4.61

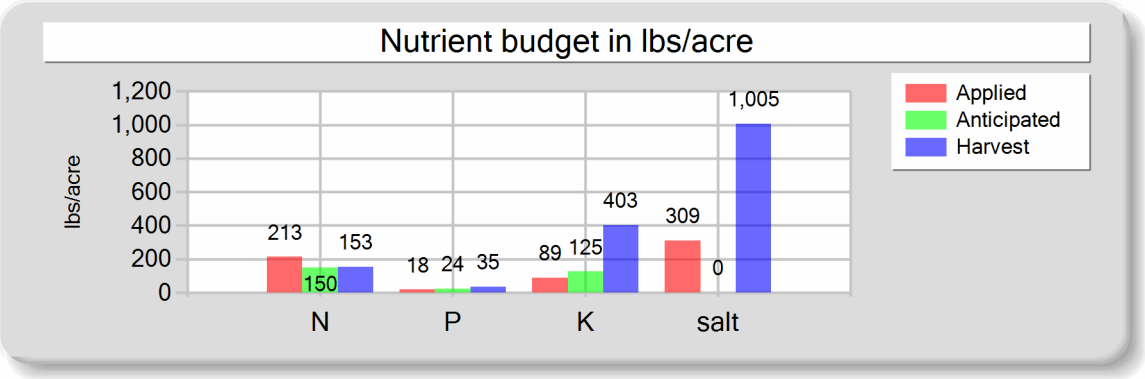
Fresh water applied	
20,790,000.00	*gallons*
765.63	*acre-inches*
45.04	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
2.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 6-7 - 11/06/2022: Oats, silage-soft dough

Field name: field 6-7 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.73	17.96	89.19	263.99
Fresh water	0.00	0.00	0.00	45.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.73	17.96	89.19	309.21
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	153.40	34.91	402.81	1,004.93
Nutrient balance	59.32	-16.95	-313.62	-695.72
Applied to removed ratio	1.39	0.51	0.22	0.31

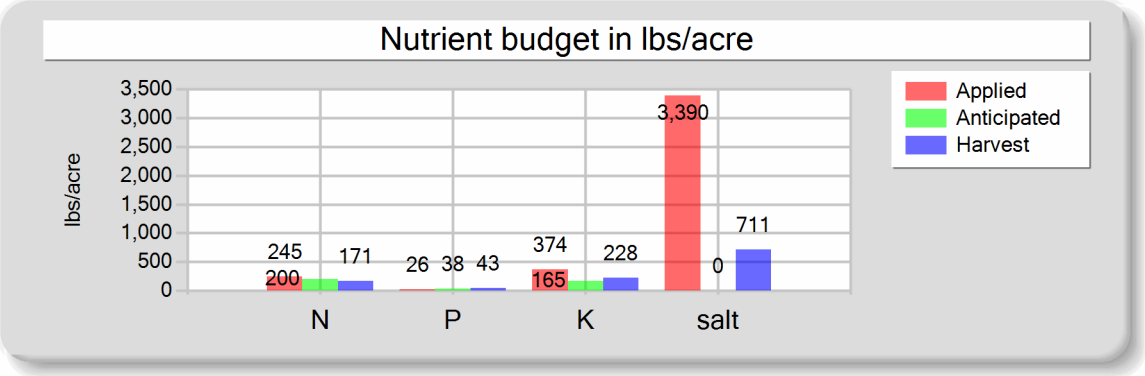
Fresh water applied	
6,075,000.00	*gallons*
223.72	*acre-inches*
6.05	*inches/acre*

Process wastewater applied	
527,250.00	*gallons*
19.42	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 6-7 - 06/15/2023: Corn, silage

Field name: field 6-7 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.37	17.54	357.28	3,098.12
Fresh water	0.00	0.00	0.00	292.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.37	26.34	373.88	3,390.21
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	171.38	42.75	228.01	711.44
Nutrient balance	73.99	-16.41	145.87	2,678.76
Applied to removed ratio	1.43	0.62	1.64	4.77

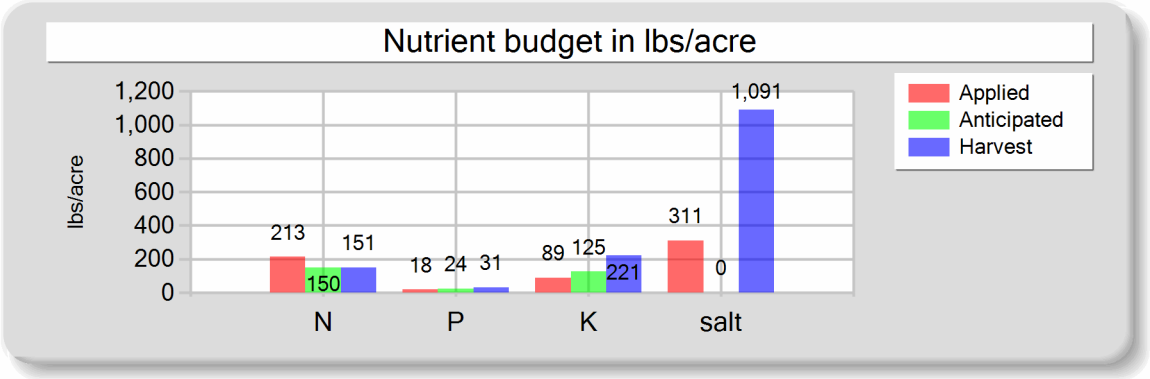
Fresh water applied	
39,244,500.00	*gallons*
1,445.24	*acre-inches*
39.06	*inches/acre*

Process wastewater applied	
2,250,000.00	*gallons*
82.86	*acre-inches*
2.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 8 - 11/07/2022: Oats, silage-soft dough

Field name: field 8 Crop: Oats, silage-soft dough Plant date: 11/07/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.73	17.96	89.19	263.99
Fresh water	0.00	0.00	0.00	47.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.73	17.96	89.19	311.08
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	150.75	30.85	221.05	1,090.52
Nutrient balance	61.97	-12.90	-131.87	-779.43
Applied to removed ratio	1.41	0.58	0.40	0.29

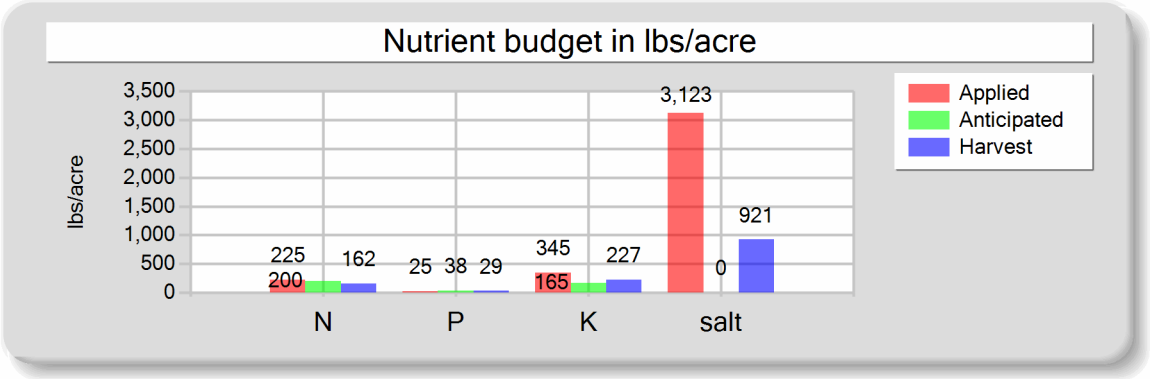
Fresh water applied	
6,156,000.00	*gallons*
226.70	*acre-inches*
6.30	*inches/acre*

Process wastewater applied	
513,000.00	*gallons*
18.89	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 8 - 06/14/2023: Corn, silage

Field name: field 8 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.26	16.26	328.13	2,819.27
Fresh water	0.00	0.00	0.00	303.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.26	25.06	344.73	3,122.89
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	162.08	28.90	226.91	921.09
Nutrient balance	63.18	-3.85	117.81	2,201.80
Applied to removed ratio	1.39	0.87	1.52	3.39

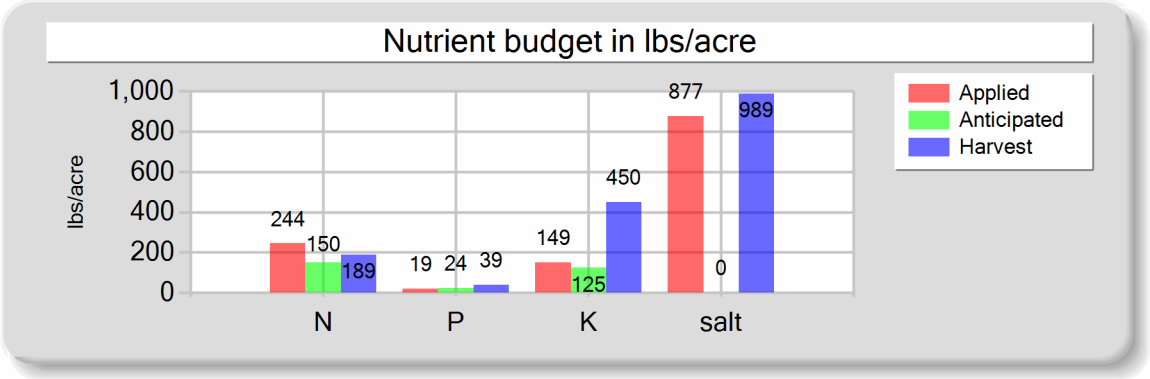
Fresh water applied	
39,690,000.00	*gallons*
1,461.65	*acre-inches*
40.60	*inches/acre*

Process wastewater applied	
2,016,000.00	*gallons*
74.24	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 9 - 11/06/2022: Oats, silage-soft dough

Field name: field 9 Crop: Oats, silage-soft dough Plant date: 11/06/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	237.33	19.49	149.41	847.18
Fresh water	0.00	0.00	0.00	29.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.33	19.49	149.41	876.93
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	188.83	38.54	450.08	989.04
Nutrient balance	55.51	-19.05	-300.67	-112.11
Applied to removed ratio	1.29	0.51	0.33	0.89

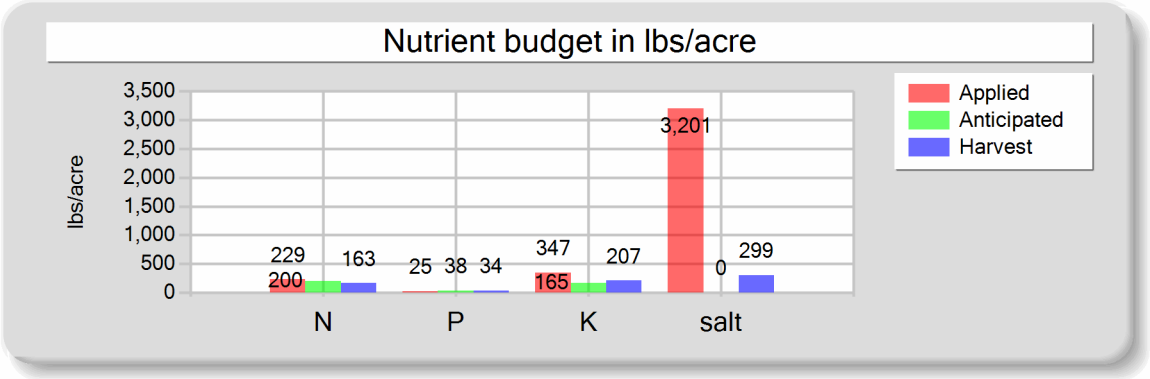
Fresh water applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
9.28	*inches/acre*

Process wastewater applied	
420,000.00	*gallons*
15.47	*acre-inches*
1.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

field 9 - 06/29/2023: Corn, silage

Field name: field 9 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	8.80	16.60	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.42	16.04	330.09	2,893.71
Fresh water	0.00	0.00	0.00	307.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.42	24.84	346.69	3,201.04
Anticipated crop nutrient removal	200.00	37.50	165.00	0.00
Actual crop nutrient removal	163.20	34.00	207.40	298.72
Nutrient balance	66.22	-9.16	139.29	2,902.32
Applied to removed ratio	1.41	0.73	1.67	10.72

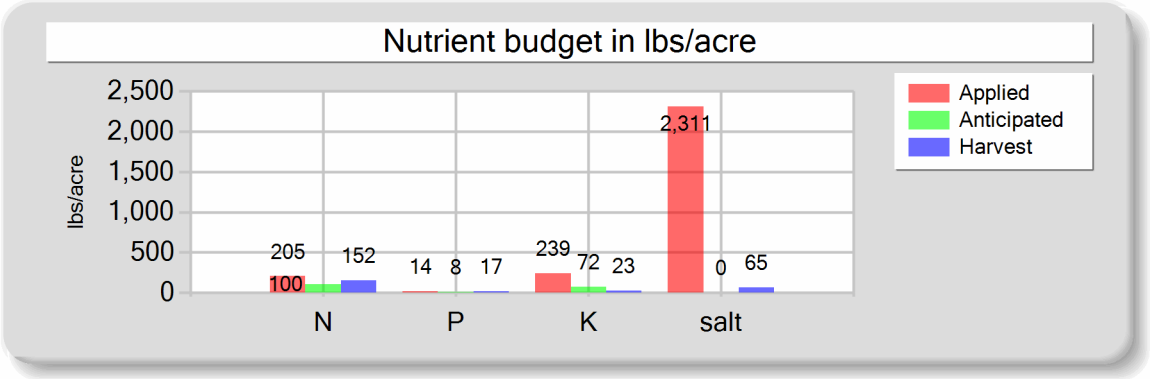
Fresh water applied	
16,740,000.00	*gallons*
616.48	*acre-inches*
41.10	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

Walnuts - 01/01/2008: Walnuts

Field name: Walnuts | Crop: Walnuts | Plant date: 01/01/2008



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.96	14.42	238.90	2,001.69
Fresh water	0.00	0.00	0.00	308.98
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	204.96	14.42	238.90	2,310.67
Anticipated crop nutrient removal	100.00	7.50	72.00	0.00
Actual crop nutrient removal	152.26	17.43	22.80	65.30
Nutrient balance	52.71	-3.01	216.10	2,245.37
Applied to removed ratio	1.35	0.83	10.48	35.39

Fresh water applied	
46,359,000.00	*gallons*
1,707.24	*acre-inches*
47.42	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/13/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,200.00	2,740.00	6,550.00							11.80
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,400.00	9,190.00	38,100.00							55.90
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 05/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	1,890.00	5,860.00							10.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 13.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	6,430.00	32,000.00							62.20
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	3,390.00	7,520.00							18.50
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 10/20/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,730.00	473.00	0.00	0.00	151.00	750.00								5,490.00	2,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/20/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	540.00	288.00	0.00	0.00	57.50	663.00								5,990.00	3,540
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ1

Sample and source description: WWQ1

Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,650.00	269.00	0.00	0.00	118.00	608.00								5,130.00	3,740
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 06/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	610.00	358.00	0.00	0.00	26.30	778.00	176.00	115.00	318.00	3,600.00	0.00	262.00	339.00	7,160.00	8,900
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	361.00	277.00	0.00	0.00	39.50	660.00								5,480.00	4,440
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	706.00	311.00	0.00	0.00	28.20	617.00								5,720.00	4,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Dairy Deep Well

Dairy Deep Well

Sample description: Dairy Deep Well

Sample date: 10/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	52.90	0.00	50.10								920.00	590
DL	0.50	0.50	0.50								1.00	10

Ranch Deep Well

Ranch Deep Well

Sample description: Ranch Deep Well

Sample date: 10/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.28	0.00	1.48								209.00	100
DL	0.50	0.50	0.50								1.00	10

TID Highline Canal

TID Highline Canal

TID Highline Canal

Sample description: TID Highline Canal

Sample date: 07/22/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.50	0.50	0.50								1.00	10

TID Highline

Sample description: TID Highline

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.20	33
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

field 10 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,230.00	1,710.00	21,100.00		9.12
DL	100.00	100.00	100.00		0.01

field 10 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,950.00	910.00	7,380.00		6.40
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,780.00	960.00	6,090.00		2.30
DL	100.00	100.00	100.00		0.01

field 11-12 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,860.00	1,110.00	10,100.00		7.70
DL	100.00	100.00	100.00		0.01

field 11-12 - 06/14/2023: Corn, silage

field 11-12 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,890.00	1,020.00	3,840.00		63.30
DL	100.00	100.00	100.00		0.01

field 13 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,440.00	1,600.00	17,300.00		10.90
DL	100.00	100.00	100.00		0.01

field 13 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,190.00	950.00	7,540.00		8.03
DL	100.00	100.00	100.00		0.01

field 14 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,020.00	1,500.00	17,300.00		12.00
DL	100.00	100.00	100.00		0.01

field 2 - 11/08/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,380.00	820.00	5,100.00		9.52
DL	100.00	100.00	100.00		0.01

field 2 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,830.00	1,020.00	5,530.00		5.90
DL	100.00	100.00	100.00		0.01

field 3 - 12/18/2017: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 10/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	1,700.00	15,900.00		11.70
DL	100.00	100.00	100.00		0.01

field 4 - 11/04/2021: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	1,650.00	16,500.00		15.50
DL	100.00	100.00	100.00		0.01

field 5 - 10/22/2018: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 11/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,100.00	1,810.00	17,500.00		11.70
DL	100.00	100.00	100.00		0.01

field 6-7 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,570.00	1,040.00	12,000.00		9.24
DL	100.00	100.00	100.00		0.01

field 6-7 - 06/15/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,690.00	1,170.00	6,240.00		5.90
DL	100.00	100.00	100.00		0.01

field 8 - 11/07/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,860.00	790.00	5,660.00		8.07
DL	100.00	100.00	100.00		0.01

field 8 - 06/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,150.00	740.00	5,810.00		8.80
DL	100.00	100.00	100.00		0.01

field 9 - 11/06/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,300.00	1,490.00	17,400.00		12.10
DL	100.00	100.00	100.00		0.01

Walnuts - 01/01/2008: Walnuts

Walnuts

Sample and source description: Walnuts

Sample date: 12/14/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,600.00	4,420.00	5,780.00		1.78
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

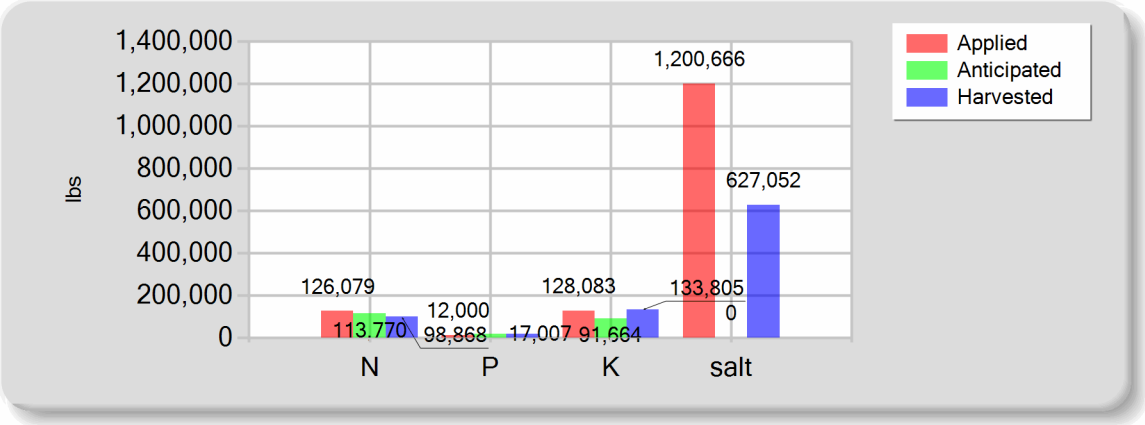
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

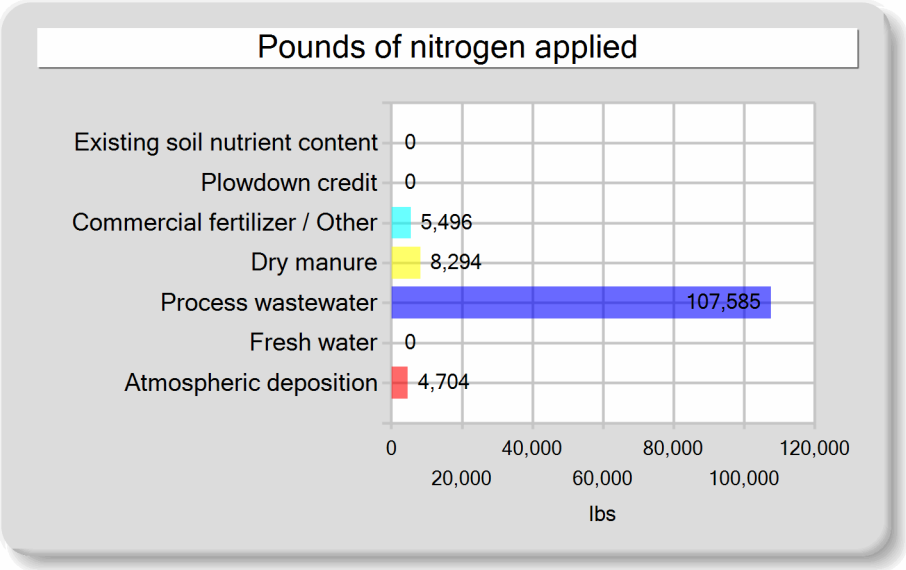
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

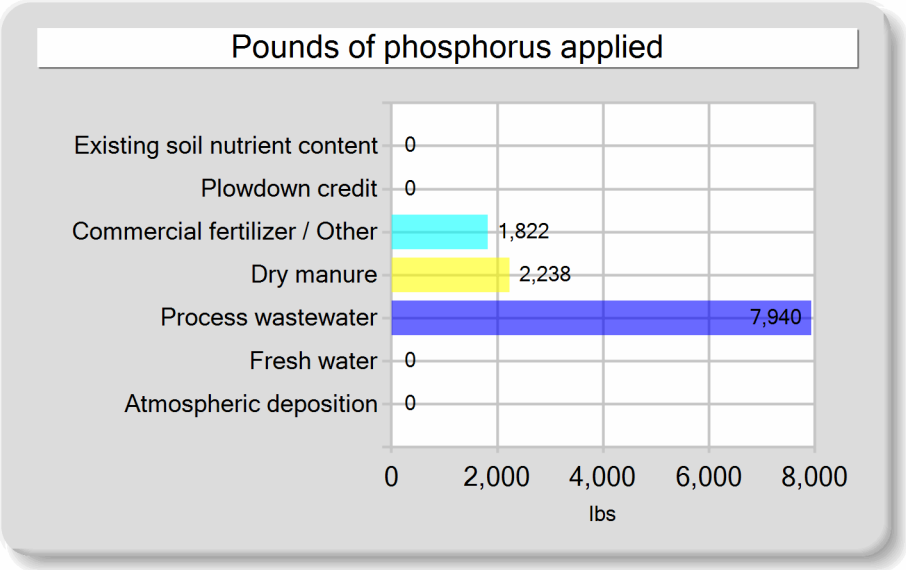
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	5,496.00	1,821.60	3,436.20	0.00
Dry manure	8,293.98	2,238.43	7,960.38	122,803.04
Process wastewater	107,585.08	7,939.51	116,686.18	983,405.78
Fresh water	0.00	0.00	0.00	94,457.40
Atmospheric deposition	4,704.00	0.00	0.00	0.00
Total nutrients applied	126,079.07	11,999.54	128,082.75	1,200,666.22
Anticipated crop nutrient removal	113,770.00	17,007.10	91,663.50	0.00
Actual crop nutrient removal	98,867.60	17,230.76	133,804.88	627,052.46
Nutrient balance	27,211.47	-5,231.22	-5,722.13	573,613.76
Applied to removed ratio	1.28	0.70	0.96	1.91

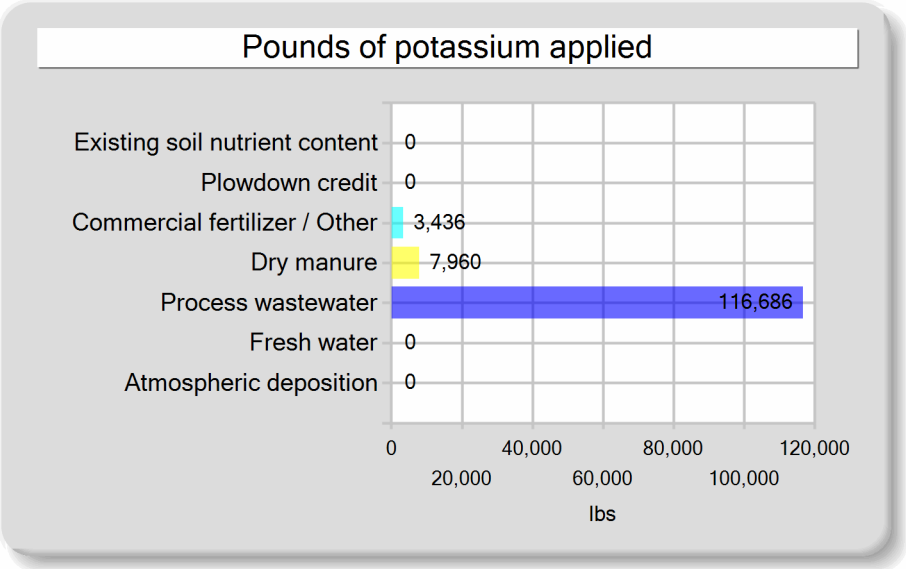
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

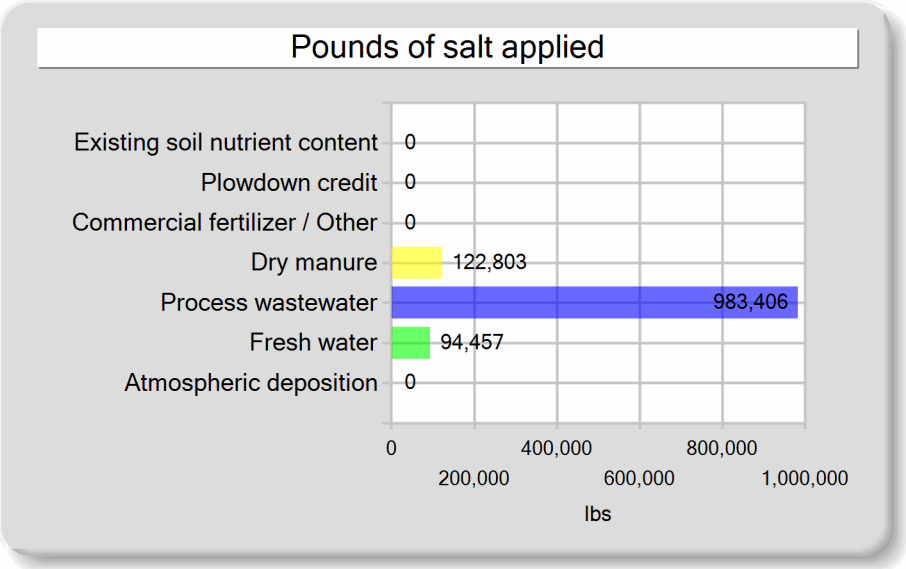


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Victor D Fiorini	Antonio Teixeira
PRINT OR TYPE NAME	PRINT OR TYPE NAME
5/31/24	5/31/24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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Compliance Analysis Report

Customer
Victor Fiorini Dairy
12926 Vincent Rd.
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ranch House	T3285039A	Electrical Conductivity (EC)	EPA 120.1 and	0.975	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	624	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)	5.64

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn Well	T3285039B	Electrical Conductivity (EC)	EPA 120.1 and	0.277	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.26	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	177	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)	5.64

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Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Barn Backup Well	T3285039C	Electrical Conductivity (EC)	EPA 120.1 and	0.975	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	624	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)	5.64

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Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Front House South 40	T3285039D	Electrical Conductivity (EC)	EPA 120.1 and	0.884	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	50.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	565	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)	5.64

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Grower

Consultant:
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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Back House South 40	T3285039E	Electrical Conductivity (EC)	EPA 120.1 and	1.26	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	64.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	804	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)	5.64

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
N. House Tony	T3285039F	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	52.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	708	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)	1.27
Electrical Conductivity (EC)	96.6			0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)	5.64

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Dairy Deep Well	T3285039G	Electrical Conductivity (EC)	EPA 120.1 and	0.920	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	50.1	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	590	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	52.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	589	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)			1.27
Electrical Conductivity (EC)	96.6			0.490 (T3284025A)	0.490 (Dup)	0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)			5.64
Total Dissolved Solids (TDS)	101			590 (T3285039G)	590 (Dup)	0.000
Total Nitrogen (TN)	101			2.80 (T3275031A)	2.80 (null)	0.000

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Compliance Analysis Report

Customer
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12926 Vincent Rd.
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: 3 A Teixeira
RECEIVED DATE: 10/12/23 4:00 PM
SUBMITTED BY: Sean
ANALYZED DATE: 10/12/23

Irrigation / Ground Water (H7)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 16.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ranch Deep Well	T3285039H	Electrical Conductivity (EC)	EPA 120.1 and	0.209	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.48	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	100	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	4.28	mg/L
		Soluble Salts (SALT-SOL)	Calculation	134	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.6	93.4 (T3284015A)	94.6 (T3284015A)			1.27
Electrical Conductivity (EC)	96.6			0.490 (T3284025A)	0.490 (Dup)	0.102
Nitrate Nitrogen (NO3-N)	100	79.8 (T3285039A)	75.5 (T3285039A)			5.64
Total Dissolved Solids (TDS)	101			590 (T3285039G)	590 (Dup)	0.000
Total Nitrogen (TN)	101			2.80 (T3275031A)	2.80 (null)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.
865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: VICTOR FIORINI
Mailing address:
(not facility address) TURLOCK CA
Phone#:

Authorized Copy Released To: F&R AG SERVICES

Sampled By: Sean Locke
Project ID: 3A TEIXEIRA

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	10/11/23	9:00	RANCH HOUSE	DOMESTIC	15.8		X				
2		9:10	MILK BARN WELL		11.8		X				
3		9:20	BARN BACKUP WELL		13.0		X				
4		9:30	FRONT HOUSE SOUTH 40		13.0		X				
5		9:40	BACK HOUSE SOUTH 40		13.0		X				
6		9:50	N. HOUSE TONY		14.0		X				
7		10:00	DAIRY DEEP WELL	AG WELL	13.0						H7
8		10:10	RANCH DEEP WELL	"	16.6						H7
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
		F&R AG SERVICES				
2nd		DENELE ANALYTICAL, INC.	10-12-23 4:00			T3285039

Required to be submitted to any agency? (YES / NO)